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Measurement of Consciousness: Subjective vs Objective

Consciousness, long considered an enigmatic phenomenon, has been the subject of philosophical speculation for centuries. However, with advancements in psychology and neuroscience, the once purely philosophical discussions have made way for empirical research in both subjective and objective measurement. At its core, consciousness encompasses the experience of awareness and the ability to have subjective experiences. Measuring such an abstract concept is inherently challenging, but several approaches have been developed in contemporary psychology. The idea that an individual's cognitive range can be meaningfully classified or even measured as conscious or unconscious has a longstanding history in modern academia and culture. In spaces such as philosophy, psychology, neuroscience, and cognitive science the term is exercised in a standard and consistent means, however all in a range of fashion unique to the subject base. In cultural, social, and medial spaces, the terms takes on less form fitting detailing, encapsulating anything from The Matrix theories to small oversights with recollection. In this paper, the alternative approaches to measurements of consciousness, both subjective and objective will be detailed, along with their criticism and limitations.

There are many methods a researcher might empirically use to measure for indicators of conscious awareness in the study of unconscious perception. Even still there exists the risk of small factorial levels of consciousness processing that researchers cannot detect and are labeling unconsciously. It’s critical to have some measure of stimulus awareness to examine a total lack of consciousness. For this, there are two major method research alternatives, Subjective and Objective approaches to the measure of consciousness research. In a subjective constructive approach, one might want to run a study based on tasks which utilize subliminal stimulus conditions by reducing stimulus intensity and incorporating masking techniques. Followed swiftly by a reduction then increase in intensity until the participant has a hard time perceiving the stimuli. Whether it be an image, characters, etc. Comparing the experimental priming tasks as conscious to the unconscious masking tasks results is assumed to offer an alright measure of consciousness. In results, unconscious effects may exist. To become more precise and

demonstrate the nature of these effects a vast measure of studies rely on the dissociation paradigm. The basic logic of such a method being that perception without awareness can be demonstrated by dissociation between two indices of perceptual processing: to indicate the availability of stimulus information to awareness consciousness and to indicate the information availability independent of whether or not the information is available to consciousness. For subliminal conditions, the goal would be to show indirect stimulus effects whilst demonstrating the participant can not perceive the stimuli in unmanipulated trials.

Subjective approaches are quite intuitive and appeal to the subjective nature of awareness. It follows the logic of collecting subjective measures of a stimulus per participant by simply asking them how much they perceived in each trial. Subjective approaches to measuring consciousness typically involve self-reporting methods, where individuals describe their own experiences, thoughts, and feelings. This approach was primarily discovered and stood as the leading method for several decades of modern consciousness research. An illustration of the method (Sidis, 1898) provided an excellent picture of the advantages of this approach. Sidis showed observers a card with dimmed lighting. Whilst the participants reported an inability to see the stimuli through the faint lighting, they were able to name the card character considerably more often than would be expected due to random guessing. On the basis of these findings Sidis concluded that his experiment highlighted a secondary subwalking self that perceives things which the primary walking self is unable to access. Other investigators (Miller, 1939) (Stroh, Shaw, and Washburn 1908) (Williams, 1938) have explored similar methods and have concluded with findings of similar strength. The advantages to this method is quite clear. As individualistic beings, it is to a degree, natural to estimate consciousness on a subjective scale. Doing so affords value for the edge cases of consciousness that would be mislabeled in less subjective methodologies of measurement.

Despite their usefulness in capturing the phenomenological aspects of consciousness, such approaches are not without criticism. There exists introspective variability with each trial, varying accessibility of mental states, a glaring lack of verifiability, retrospective distortion and memory bias, language limitations, and reliance on conscious awareness within the subjective approach that highlight a few shortcomings. Despite these criticisms, subjective measures continue to provide valuable insights into the phenomenological aspects of consciousness that objective measures might overlook or be unable to ascertain.

A majority of the criticism of subjective approaches stem from the discoveries in the advancement of Signal Detection Theory (SDT). While subjective reports were once predominant, SDT has revealed that an

observer's discriminatory responses are not direct reflections of their perceptual experience but rather the combined result of distinct perceptual and decision-making processes. SDT has been instrumental in demonstrating how discriminatory responses can be decomposed into separate components of perceptual sensitivity and decision criteria. Essentially, responses do not provide a transparent window into inner experiences or awareness. Instead, perception is not all-or-nothing but continuous, influencing the gradation of our awareness of a stimulus. Responses rely on a criterion that gauges the sufficiency of perceived evidence, meaning that claims of no awareness do not necessarily indicate an absence of awareness, but that it might simply fall below the established threshold. The insights from SDT are crucial for discerning conscious from unconscious processes.

SDT virtually stopped unconscious perception research's advantage in the field. In these ways and more, SDT has fundamentally altered our understanding of perceptual and memorial processes, as well as decision processes generally. Despite the overwhelming evidence favoring the SDT framework over classical subjective threshold models, many continue to question perceptual sensitivity's relevance to consciousness. These concerns largely arise from the mistaken belief that perceptual sensitivity is merely behavioral and disconnected from subjective experience. Contrary to this view, an in-depth look at the SDT rating task indicates that perceptual sensitivity more accurately reflects subjective confidence within a framework that separates the ability to discriminate between stimulus classes based on confidence from the criteria we use for responding. This distinction prevents the conflation of these two distinct processes. A critical insight might be gained by examining the qualitative discrepancies between the impacts of stimuli that are subject to unconscious versus overt conscious perception. There are many defenses to subjective approaches, even after criticism. These include that the method emphasizes intuitive appeal, although this doesn't address SDT directly. Arguments that typical above-chance performance found subjectively is indeed unconscious and that qualities of consciousness effects differ in more than quantitative ways, but qualitatively also. It introduces the concept of conscious control and automatic responses.

With the understanding of the importance of SDT, objective approaches were developed to mitigate the downfalls of subjective methods. It's important to understanding the nature of objective approaches that one remembers they developed in response to a critique of subjective approaches and encouraged the use of discrimination performance

Objective methods that use discrimination performance refer to the techniques and tests employed in psychological research and assessment to measure an individual's ability to distinguish between different stimuli.

The approach argues that typical above-chance performance under subjective conditions could plausibly reflect weak, faint, or indistinct results but nonetheless conscious perception. To rule out this possibility of perceptual sensitivity of zero, there's a requirement for far more stringent stimulus conditions. Then if one still gets evidence of cognitive processing, one can conclude such effects are truly a reflection of zero stimuli unconsciousness. These methods are considered objective because they rely on observable and quantifiable behaviors rather than subjective self-reports of experiences or sensations. The goal of using discrimination performance as a metric is to infer cognitive processes such as those involved in attention, perception, memory, or consciousness.

Objective Approaches to measuring consciousness range in methods. Examples of discrimination tasks include SDT, hence the reason it served to open the research community to subjective method flaws. Additional discrimination performance tasks include Visual Discrimination Tasks where participants might be asked to differentiate between images or patterns that are presented visually. The ability to discern slight differences in shape, color, or spatial orientation, for example, can serve as an objective measure of visual processing. Auditory Discrimination Tasks where similarly these assess an individual's ability to discriminate between different sounds, such as varying pitches, tones, or speech sounds. Performance can indicate auditory processing capabilities and attentional focus. Somatosensory Discrimination Tasks include differentiating between tactile stimuli, such as textures or points of pressure. They often assess the sensory and perceptual accuracy of an individual's somatosensory system. Discrimination performance is typically analyzed by looking at accuracy, reaction time, and sensitivity. The objectivity of these methods rests in the premise that the results do not solely depend on subjective reporting but can be quantified and statistically analyzed. They allow researchers to draw conclusions about cognitive functions that underlie the discrimination performance, such as conscious perception or attention, and can be particularly useful in clinical settings where the patient might not be able to communicate their experiences directly due to impairment or unconsciousness. Such objective methods remain indispensable tools in psychology for understanding a range of cognitive processes in both healthy individuals and clinical populations.

One of the most prominent tools for objectively measuring consciousness is neuroimaging, including functional magnetic resonance imaging (fMRI) and positron emission tomography (PET) scans. These techniques measure brain activity by detecting changes in blood flow or metabolic processes, making it possible to identify patterns of neural activation that correlate with conscious experiences. For instance, certain neural signatures are associated with the state of being awake and aware, as opposed to when one is unconscious or under anesthesia.

Techniques such as electroencephalography (EEG) and magnetoencephalography (MEG) record electrical or magnetic activity in the brain, capturing real-time fluctuations that reflect the dynamic changes in consciousness. Event-Related Potentials (ERPs), derived from EEG, track how the brain responds to specific stimuli, revealing the various stages of sensory processing and indicating at what point consciousness of a stimulus arises. While not directly probing the brain, behavioral tests can infer conscious experience from the predictable performance of certain tasks. For example, the use of signal detection tasks, where participants must detect a faint stimulus presented briefly, can reveal thresholds for conscious detection. Furthermore, tasks that probe attention, working memory, and executive control can also serve as indirect measures of consciousness. Based on The Global Workspace Theory, for a piece of information to become conscious, it must be broadcast throughout the brain via a global workspace. Tests that measure the integration and broadcasting of information, such as the extent of network connectivity in the brain, can inform assessments of consciousness.

Criticisms in this method include the argument that the use of discrimination performance doesn't target awareness and instead index objective behavior. This criticism ignores the truth that objective approaches measure subject reported confidence. Despite these advances, it's imperative to note that all objective measures are ultimately proxies for the subjective quality of consciousness and may not capture its entirety. The results are extremely faint and it takes much editing of recorded tasks (deletion of noise, averaging of trials, etc) to render valuable results. These results, however, will be backed quantitatively if not subject to bias. Such measurements are proxies for consciousness, providing a practical and empirical approach while acknowledging the subjective and intrinsic nature of conscious experience. As research technologies and theoretical models evolve, objective measurements of consciousness in psychology continue to refine and challenge our understanding. These efforts bridge the divide between the subjective and objective realms, paving the way for a deeper comprehension of the human mind.

Certain elements must be satisfied before an adequate measure is analyzed. A sufficient measure of the perceptual information must be selected. This measure of perceptual experience must show null sensitivity. And, given this sensitivity is measured, the second measure of processing must be shown to have greater than zero sensitivity. These encompass the three requirements and consequently limitations on the measurement of consciousness: Exhaustiveness, Exclusiveness, and Null Sensitivity.

The exhaustiveness problem could refer to the difficulty in achieving a complete measurement of consciousness, encompassing all its facets, whether subjective or objective, although its solution is more object

oriented. Some might suggest it is a subjective problem. The measurement of consciousness is a complex task because consciousness is inherently subjective, leading to difficulties in obtaining exhaustive, objective measures. There are several problems and distinctions to consider, including subjective versus objective aspects. It's difficult if not impossible to know if a subjective report covers the full range of an individual's conscious experience. Different people may have varying abilities to introspect and articulate their experiences, and some aspects of consciousness may remain private or unreportable. Therefore, achieving exhaustiveness in subjective reports is a significant challenge. This issue is also linked with the problem of other minds, which is about our inability to directly access the conscious experiences of others. No objective measure can capture the full complexity of subjective experience. While we may correlate certain neural patterns or behaviors with conscious states, we cannot be sure these measurements account for the entirety of what it is like to have those experiences from the inside (the qualia). There's always the possibility that some dimensions of consciousness elude these objective measures, making the task of finding exhaustive objective measures of consciousness very difficult. The challenge is compounded by what's often referred to as the hard problem of consciousness, which centers on the difficulty of explaining why and how physical processes in the brain give rise to subjective experience. Addressing these problems involves a multi-pronged approach that can include developing and refining psychophysical methods to better relate subjective reports to objective data. Using neuroimaging and neurophysiological techniques to map correlates of consciousness in the brain is one example. Investigating altered states of consciousness to gain insights into different levels and aspects of consciousness, another. Philosophical analysis to clarify the conceptual underpinnings of consciousness and its measurement, a final. Consciousness is a multifaceted construct that is not easily captured in its entirety by any one measurement approach.

The exclusiveness problem in the context of measurement of consciousness refers to the challenge of ensuring that the measures used to assess consciousness are specific to consciousness itself and not confounded by other cognitive or behavioral factors. This is a significant hurdle in both subjective and objective approaches to studying consciousness. When measuring consciousness, the goal is to find measures that are sensitive to variations in consciousness and are not simply indicative of other unrelated processes. Here's how the exclusiveness problem can present itself in both subjective and objective measures. When relying on subjective reports, researchers must determine whether those reports are truly reflective of consciousness or whether they might be influenced by other factors, such as language abilities, desire to please the experimenter, or even the individual's interpretation of the

task. There's a risk that what's being measured could be an individual's communicative capabilities or their understanding of the question rather than their conscious experience. The challenge here is ensuring that subjective reports uniquely reflect consciousness without being contaminated by these other cognitive processes or biases. Objective measures, such as brain activity patterns, physiological responses, and behavioral tests, aim to bypass the need for subjective reporting. However, they too can be affected by factors unrelated to consciousness. For instance, a particular neural signal may correlate with attention or memory processing rather than consciousness itself. The difficulty lies in isolating measurements that are exclusive indicators of conscious experience and are not reflective of unconscious processing or other cognitive functions. Addressing the exclusiveness problem involves careful experimental design and the development of new theories and methods that can better differentiate between conscious and non-conscious processes. Neuroscientists and psychologists employ various strategies to tackle this issue including Consciousness Contrasts, SDT, Neural Correlates of Consciousness, and No-Report Paradigms.

The null sensitivity problem in the measurement of consciousness refers to the issue of whether an empirical measure is sensitive enough to detect the presence or absence of consciousness. This problem can occur in both subjective and objective assessments of consciousness and is particularly pertinent when trying to determine levels of consciousness in individuals who might not be able to communicate their experiences, such as patients with disorders of consciousness or in cases where consciousness is not readily apparent, such as in certain animal species or artificial systems. Subjective measures depend on individual self-report and introspection. The null sensitivity problem here could relate to whether the subjective report or scale used is fine-grained enough to pick up on the nuances of conscious experience. In subjective reporting, an individual may not report any conscious experience either because the subjective measure isn't sensitive enough to detect low-intensity experiences or because there is genuinely no experience to report. This creates a dilemma in distinguishing between these possibilities. Objective measures, however, rely on physiological or behavioral indicators to infer consciousness. The null sensitivity problem here refers to whether these measures can detect the presence of consciousness, especially in cases where it might be minimal or different from the norm. In objective assessments neurological tests or behavioral assessments might fail to detect consciousness, suggesting it is absent when, in fact, it might exist but not manifest in observable ways due to the limitations of the test or measurement tools employed. The null sensitivity problem is an ongoing challenge in consciousness research. It demands continuous efforts to verify that measurement tools are not only accurately reflecting consciousness but also that they are sufficiently sensitive to detect instances where

consciousness might be present, even if at a minimal level or in a form that is different from the typical human experience.

The issue of subjective and objective incompatibility in the measurement of consciousness centers around the concern that these two approaches might not align because they could be indexing different aspects or dimensions of consciousness. This presents a significant challenge for researchers who aim to develop an integrative and comprehensive understanding of consciousness. This incompatibility is birthed from several issues. One being different domains of measurement as subjective measures capture phenomenological aspects—what it feels like from the inside to be conscious—while objective measures capture physical processes that are associated with, but not necessarily identical to, those subjective experiences. Also communication limitations as subjective reports can be influenced by language and cognitive biases. An individual's ability to communicate their conscious experiences can vary, leading to discrepancies that do not accurately reflect underlying conscious states. Additionally, conscious versus unconscious processing as not all cognitive processes are conscious, and objective measures may detect unconscious processes. Therefore, objective data can sometimes reflect operations that occur without subjective awareness, and this can complicate the interpretation of such data in terms of consciousness.

The conceptual discrepancy between subjective and objective approaches may indeed reflect their indexing of fundamentally different processes. There are a multitude of alternative frameworks that exist in an attempt to put this issue into perspective. Snodgrass (2004) advocates for a novel methodological framework that emphasizes the necessity of defining a single process model for consciousness perception. It endorses a hierarchical model of conscious perception based on strength and complexity. SDT implies a hierarchical functioning of conscious perception, suggesting that higher-order effects, such as semantic priming, cannot occur without the foundation of lower-level effects like detection and identification. Yet, paradoxically, a review of the literature reveals that direct and indirect measures often share a non-linear or even negative relationship, rather than the predicted positive one. This not only furnishes potent evidence for the existence of unconscious perception but also implies that direct, subjective measures may overshadow indirect, objective ones when both are simultaneously engaged, obscuring the full picture of how consciousness operates. I argue that combining subjective with objective methods is often considered the most informative approach when studying the multifaceted nature of consciousness in psychology. The focus should be on bridging methodologies to implement subjective factors into objective tests.

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